AI-DRIVEN VIRTUAL MEMORY OPTIMIZER AND HARDWARE SIMULATOR

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1. INTRODUCTION

Modern computational infrastructure running intensive Artificial Intelligence (AI) and Deep Learning (DL) workloads faces a structural performance bottleneck known as the "Memory Wall." As neural network models scale exponentially to billions of parameters, storing expansive multi- dimensional tensor layers entirely within high-speed, premium local DRAM becomes cost- prohibitive for commodity data centers. Consequently, modern server infrastructure is transitioning toward Heterogeneous Memory Architectures (HMAs). These frameworks interface a limited pool of ultra-fast local DRAM with larger, cost-effective, highlatency storage tiers like Compute Express Link (CXL)-attached Persistent Memory or NonVolatile Memory (NVM). While this hardware alignment expands memory volume, data management across these asymmetric performance tiers is traditionally delegated to general purpose Operating Systems (OS) using demand-paging heuristics like Least Recently Used (LRU). While effective for standard software workloads, LRU suffers severe performance degradation under deep learning applications. The highly cyclic, predictable, yet non-linear looping structures of matrix multiplication algorithms break temporal locality, causing severe page thrashing and interconnect bus saturation. This proposed project implements a highperformance software-hardware simulation workbench to resolve this data routing bottleneck. By designing a trace-driven core that models virtual-to-physical address translation layers, our project extracts fine-grained spatial-temporal access telemetry. Rather than applying legacy reactive page swaps, this system introduces an intelligent, proactive predictive preemption layer. This framework aims to bridge fundamental systems concepts taught in Operating Systems and Computer Architecture with modern deployment requirements, validating theoretical latency optimization rules within specialized cloud data configurations.

2. LITERATURE SURVEY

Contemporary research highlights significant challenges in orchestrating heterogeneous memory configurations. Standard computing baselines utilize native operating system frameworks like Linux AutoNUMA and numad. These utilities rely heavily on hardware page-fault tracking

cycles to balance multi-socket data allocations across local versus remote nodes. However, these systems operate reactively; a execution thread must endure the full processing penalty of an address translation failure or physical page fault before the kernel initiates data migration routines. Recent system literature describes attempts to mitigate these delays by hardcoding software prefetching modules or optimizing page size allocations using Transparent Huge Pages (THP). While these techniques reduce internal page table page walks, they lack dynamic workload context, frequently causing redundant page caching and bus congestion under complex tensor workloads. The primary gap identified in existing literature is the heavy reliance on historical heuristics for workloads that possess predictable forward-looking structures. Deep learning models execute deterministic computational steps, yet native general-purpose operating systems treat memory streams as randomized inputs. This project addresses this gap by replacing history-bound page replacement algorithms with a forward-looking, learning-assisted policy. By profiling access metrics using microsecond-scale execution features, our framework moves beyond the rigid constraints of native OS balancing systems. It provides a specialized, domain-specific optimization space tailored for dedicated neural network architectures, proving that predictive preemption can bypass the performance trade-offs of reactive general-purpose operating systems.

3. PROBLEM DOMAIN

The primary problem addressed by this project is the severe latency penalty and memory bus thrashing caused by the inability of traditional Operating System page-replacement policies (such as LRU) to manage asymmetric memory tiers during AI tensor operations. Because deep learning algorithms read and write data in massive, cyclic looping arrays, legacy caching schemes consistently evict critical address spaces prematurely. This leads to continuous synchronous page fetches from high-latency NVM devices, which drastically inflates the Effective Memory Access Time (EMAT), induces heavy tail latency spikes, and wastes processor execution cycles.

Objectives of the Proposed Project:

To engineer a high-speed, trace-driven simulation environment in C++ that replicates multi- level hardware memory translations including TLB caches, Multi-level Page Tables, DRAM, and NVM pools.

To build an intelligent, lightweight machine learning classifier capable of predicting optimal page placement with minimal microsecond-scale computational overhead.

To design a web-based, interactive visual utility showing live page mappings and performance analytics to satisfy university application specifications.

4. SOLUTION DOMAIN

[Workload Data Log] → [C++ Hardware Engine] → [AI Predictor] → [Streamlit Panel]

The proposed solution implements a modular software-hardware co-design simulation architecture divided into four functional stages: workload data tracking, architectural hardware

state modeling, machine learning classification, and visual telemetry reporting. First, a Python-based workload generation layer runs simulated deep learning operations (such as dense tensor multi-array operations), generating an execution log mapping exact 64-bit virtual memory hex addresses alongside operation markers (Read/Write) and microseconds elapsed. Second, the system core features a high-performance simulation engine compiled in pure C++17. This engine parses the log stream line-by-line. It uses low-level bit-shifting operations to isolate the Virtual Page Number (VPN) from the offset elements (VPN = Address >> 12). The engine maintains independent state trackers modeling a 16-entry Translation Lookaside Buffer (TLB), an O(1) unordered page table tracking map, and a physical DRAM allocation bank to compute baseline performance. Third, instead of relying on legacy LRU rules, the allocation choices are handed to a trained Random Forest Classifier built using Python's scikit-learn framework. This model takes low-overhead features extracted by the simulator— specifically page access frequency, reuse distance, and read/write configurations—and outputs a binary prediction (1 for DRAM promotion, 0 for NVM demotion) to preemptively place pages into fast frames before a processor request stalls. Finally, the frontend is built using Python Streamlit, acting as an interactive control panel for hardware configurations and live performance analytics.

5. SYSTEM DOMAIN

Hardware Specifications:

- Processor: Minimum Intel Core i3/i5 or AMD Ryzen 3/5 consumer laptop architecture.
- Memory: 8 GB RAM minimum; 16 GB high-speed DDR4/DDR5 configuration preferred.
- Storage: 10 GB free solid-state drive (SSD) space for housing multi-row trace data logs.

Software Specifications:

- Operating System: Windows 10/11 or Ubuntu Linux environments.
- Languages & Toolchains: C++17 toolchain (g++); Python 3.10+ execution environment.
- Libraries: scikit-learn, pandas, numpy, streamlit, and plotly.

6. APPLICATION DOMAIN

The primary application scope of this framework centers on Domain-Specific AI Infrastructure and Dedicated Cloud Inference Farms hosting large models. While the microsecond processing overhead of a machine learning classifier makes it impractical for general-purpose personal laptops, it fits perfectly within dedicated server clusters. Since these platforms process highly repetitive, cyclic tensor operations all day, our predictive allocation design minimizes expensive tail latencies. This framework also serves as an educational utility for computer science departments, visually demonstrating complex OS and hardware architecture mechanics during academic coursework.

7. EXPECTED OUTCOME

A fully functional, trace-driven C++ hardware engine capable of parsing thousands of references in milliseconds.

A trained, lightweight Random Forest Classifier achieving allocation prediction accuracy above 94%.

An interactive Streamlit dashboard displaying real-time EMAT changes and page-fault curves.

A performance speedup demonstrating up to a 30% reduction in EMAT over native LRU rules.

A formal LaTeX research paper draft formatted to IEEE conference standards.

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